Marking memorandum

For the tutor · keep this file until the paper has been written

One valid method is shown. An equivalent correct method earns the same marks. A correct final answer earns the marks for that part. If the final answer is wrong, award the method marks that were actually earned. Do not take the same mistake off twice: if a later part uses an earlier wrong answer correctly, award the follow-through marks described below.

Question	Topic	Marks
1	Whole numbers	**8**
2	Ratio, rate, proportion and finance	**10**
3	Integers	**6**
4	Exponents	**10**
5	Algebraic expressions	**12**
6	Factorising and algebraic fractions	**10**
7	Equations	**10**
8	Patterns, functions and graphs	**10**
9	Geometry	**10**
10	Pythagoras, circles and volume	**10**
11	Transformations	**4**
1 to 11	Timed paper	**100**
12	Data and probability (optional)	**10**

QUESTION 1 · Whole numbers · 8 marks

1.1 Choose the **most precise** description for each number. Write only the letter. **(4)**

A natural number
B whole number, but not a natural number
C integer, but not a whole number
D rational, but not an integer
E irrational

(a) 0 (b) −11 (c) 2/9 (d) $\sqrt{7}$

(a) B (b) C (c) D (d) E

One mark for each letter. In this paper, as in Platinum, the natural numbers start at 1, so 0 is a whole number and not a natural number.

1.2 Write 90 as a product of its prime factors, in exponential form. **(2)**

90 = 2 × 45 = 2 × 3 × 15 = 2 × 3 × 3 × 5 = **$2 \times 3^2 \times 5$**

2 marks for $2 \times 3^2 \times 5$. Award 1 mark for a correct product that is not yet in exponential form, such as 2 × 3 × 3 × 5.

1.3 Find the HCF and the LCM of 28 and 42. **(2)**

$28 = 2^2 \times 7$ and $42 = 2 \times 3 \times 7$.

HCF = $2 \times 7 =$ **14** (1)

LCM = $2^2 \times 3 \times 7 =$ **84** (1)

Check: $14 \times 84 = 1\,176$ and $28 \times 42 = 1\,176$.

QUESTION 2 · Ratio, rate, proportion and finance · 10 marks

2.1 A bus travels 210 km in 2 hours 30 minutes. Calculate its average speed in km/h. **(2)**

2 hours 30 minutes = 2,5 hours. (1)

Speed = 210 ÷ 2,5 = **84 km/h**. (1)

Also accept 210 km in 150 minutes, then 1,4 km/min × 60 = 84 km/h. The unit km/h is required for the second mark.

2.2 The ratio of red beads to blue beads is 5 : 3. There are 48 beads altogether. How many red beads are there? **(2)**

5 + 3 = 8 parts, so one part is 48 ÷ 8 = 6. (1)

Red beads = 5 × 6 = **30**. (1)

2.3 4 m of fabric costs R78. Calculate the cost of 7 m of the same fabric. **(2)**

This is direct proportion. Cost = 7/4 × 78. (1)

= **R136,50**. (1)

Or R78 ÷ 4 = R19,50 per metre, then R19,50 × 7 = R136,50.

2.4 It takes 6 machines 8 hours to print a batch of books. How long will 4 machines take if they work at the same rate? **(2)**

Indirect proportion. The job is 6 × 8 = 48 machine-hours. (1)

Time for 4 machines = 48 ÷ 4 = **12 hours**. (1)

More machines must take less time. An answer above 8 hours is the one to expect; an answer below 8 hours came from treating it as direct.

2.5 R4 000 is invested at 5% per annum, compounded yearly, for 2 years. Calculate the value of the investment after 2 years. **(2)**

$A = 4\,000 \times (1{,}05)^2$ or year-by-year: 4 000 × 1,05 = 4 200. (1)

$A = 4\,000 \times 1{,}1025 =$ **R4 410**. The second year is 4 200 × 1,05 = R4 410. (1)

QUESTION 3 · Integers · 6 marks

3.1 Calculate. **(6)**

3.1.1 $-8 + (-6)$ (1)
3.1.2 $4 - (-7)$ (1)
3.1.3 $(-3) \times (-4) \times (-2)$ (2)
3.1.4 $-36 \div 4 + (-2)^2$ (2)

3.1.1 $-8 + (-6) =$ **-14** (1)

3.1.2 $4 - (-7) = 4 + 7 =$ **11** (1)

3.1.3 $(-3) \times (-4) = 12$, then $12 \times (-2) =$ **-24**. One mark for 12, one mark for −24. Three negative factors give a negative answer.

3.1.4 $(-2)^2 = 4$ and $-36 \div 4 = -9$, so $-9 + 4 =$ **-5**. One mark for either correct piece, one mark for −5.

QUESTION 4 · Exponents · 10 marks

4.1 Simplify. Where a power is negative, write the answer with a positive exponent. **(9)**

4.1.1 $y^6 \times y^2$ (1)
4.1.2 $(5^2)^2$ Write the answer as a number. (1)
4.1.3 $(2p^2q)^3$ (2)
4.1.4 $3^{-2} + 7^0$ (2)
4.1.5 $(8x^6y^2) \div (2x^2y^5)$ (3)

4.1.1 $y^{6+2} =$ **y^8** (1)

4.1.2 $5^4 =$ **625** (1). Also accept 5^4 for the mark.

4.1.3 $2^3 \times p^6 \times q^3 =$ **$8p^6q^3$**. One mark for 8, one mark for p^6q^3.

4.1.4 $3^{-2} = 1/9$ and $7^0 = 1$, so the sum is **10/9**. One mark for either correct piece, one mark for 10/9. Also accept $1^1/_9$.

4.1.5 $8 \div 2 = 4$, $x^{6-2} = x^4$, and $y^{2-5} = y^{-3}$. (2 for these three pieces, at least the coefficient and one correct power)
$=$ **$4x^4 / y^3$** (1). Also accept $4x^4y^{-3}$.

4.2 Write 0,0064 in scientific notation. **(1)**

$6{,}4 \times 10^{-3}$ (1). The first number must be at least 1 and less than 10. Also accept 6.4×10^{-3}.

QUESTION 5 · Algebraic expressions · 12 marks

5.1 The expression $-5x^2 + x - 9$ is given. **(2)**

5.1.1 Write down the coefficient of x^2. (1)
5.1.2 Write down the constant term. (1)

5.1.1 **-5** (1). The minus belongs to the coefficient.

5.1.2 **-9** (1).

5.2 Simplify. **(6)**

5.2.1 $6a - 4b - a + 7b$ (1)
5.2.2 $5(x - 2) - 3(2x + 1)$ (3)
5.2.3 $(18m^2n - 12mn^2) \div 6mn$ (2)

5.2.1 **$5a + 3b$** (1)

5.2.2 $5x - 10 - 6x - 3$ (2: one mark for each bracket, including the signs)

$=$ **$-x - 13$** (1)

5.2.3 $18m^2n \div 6mn = 3m$ and $-12mn^2 \div 6mn = -2n$, so **$3m - 2n$**. One mark for $3m$, one mark for $-2n$.

5.3 Expand and simplify. **(4)**

5.3.1 $(x + 7)(x - 3)$ (2)
5.3.2 $(2x - 3)^2$ (2)

5.3.1 $x^2 - 3x + 7x - 21 =$ **$x^2 + 4x - 21$**. One mark for the four products or an equivalent expansion, one mark for the simplified form.

5.3.2 $(2x)^2 - 2(2x)(3) + 3^2 =$ **$4x^2 - 12x + 9$**. One mark for the correct middle term $-12x$, one mark for the two squares.

QUESTION 6 · Factorising and algebraic fractions · 10 marks

6.1 Factorise fully. **(8)**

6.1.1 $7x^2 - 21x$ (2)
6.1.2 $16 - p^2$ (2)
6.1.3 $x^2 + 2x - 15$ (2)
6.1.4 $2x^2 - 8x - 10$ (2)

6.1.1 **$7x(x - 3)$**. One mark for the common factor $7x$, one mark for the bracket.

6.1.2 **$(4 - p)(4 + p)$**. Also accept $(4 + p)(4 - p)$. One mark if only the two square roots are correct.

6.1.3 **$(x + 5)(x - 3)$**. The numbers 5 and -3 multiply to -15 and add to 2.

6.1.4 $2(x^2 - 4x - 5) =$ **$2(x - 5)(x + 1)$**. One mark for the common factor 2, one mark for both brackets. $(2x + 2)(x - 5)$ is not yet fully factorised.

6.2 Simplify $(x^2 - 16) / (x + 4)$. **(2)**

$(x - 4)(x + 4) / (x + 4) =$ **$x - 4$**, with **$x \neq -4$**.

One mark for $x - 4$. One mark for the restriction. The original denominator is 0 when $x = -4$.

QUESTION 7 · Equations · 10 marks

7.1 Solve for x. **(8)**

7.1.1 $4x + 3 = -13$ (2)
7.1.2 $2(x - 5) = 3x + 1$ (3)
7.1.3 $x^2 - 9x = 0$ (3)

7.1.1 $4x = -16$ (1), so **$x = -4$** (1).

7.1.2 $2x - 10 = 3x + 1$ (1), then $-10 - 1 = 3x - 2x$ (1), so **$x = -11$** (1).

Check for 7.1.2: $2(-11 - 5) = -32$ and $3(-11) + 1 = -32$.

7.1.3 $x(x - 9) = 0$ (1), so **$x = 0$** (1) or **$x = 9$** (1). Dividing by x loses the answer 0, so do not award that lost answer.

7.2 A mother is three times as old as her son. The sum of their ages is 48 years. How old is the son? **(2)**

Let the son's age be x years. Then the mother is 3x, and x + 3x = 48. (1)

4x = 48, so **x = 12**. The son is 12 years old. The mother is 36. (1)

QUESTION 8 · Patterns, functions and graphs · 10 marks

8.1 The pattern is 8 ; 13 ; 18 ; 23 ; … **(4)**

8.1.1 Write a formula for the general term, T_n. (2)

8.1.2 Which term is equal to 78? (2)

8.1.1 The difference is 5, so $T_n = 5n + c$. Using the first term, 5 + c = 8, so c = 3. **$T_n = 5n + 3$**. One mark for the 5n, one mark for + 3. Test: n = 2 gives 13.

8.1.2 5n + 3 = 78 (1), so 5n = 75 and **n = 15**. It is the 15th term. (1)

Follow through: if their rule in 8.1.1 was wrong, mark 8.1.2 from that rule and award the method.

8.2 A relationship has the rule: output = −2 × input + 6. **(3)**

8.2.1 Find the output when the input is −3. (1)

8.2.2 Find the input when the output is 0. (2)

8.2.1 −2 × (−3) + 6 = 6 + 6 = **12** (1)

8.2.2 −2x + 6 = 0 (1), so **x = 3** (1). This input is also the x-intercept of y = −2x + 6.

8.3 Determine the equation of the straight line passing through (0 ; −2) and (3 ; 4). **(3)**

m = (4 − (−2)) / (3 − 0) = 6/3 = 2. (1)

The y-intercept is −2, because the line passes through (0 ; −2). (1)

y = 2x − 2. (1). Check: when x = 3, y = 6 − 2 = 4.

QUESTION 9 · Geometry · 10 marks

Diagram reminder: a is vertically opposite 58°, b is the co-interior angle, and c is the corresponding angle.

9.1 In the diagram, AB ∥ CD and EF is a transversal. Determine, with reasons: **(6)**

9.1.1 the size of a, the angle vertically opposite the given 58° angle (2)

9.1.2 the size of b, the co-interior angle to the given 58° angle (2)

9.1.3 the size of c, the corresponding angle to the given 58° angle (2)

9.1.1 a = **58°**. Vertically opposite angles are equal. One mark for 58°, one mark for the reason. Parallel lines are not needed for this reason.

9.1.2 b = 180° − 58° = **122°**. Co-interior angles are supplementary, AB ∥ CD. One mark for 122°, one mark for the reason.

9.1.3 c = **58°**. Corresponding angles are equal, AB ∥ CD. One mark for 58°, one mark for the reason.

9.2 In △KLM, ∠K = 2x, ∠L = 3x and ∠M = 4x. Calculate the size of the largest angle. **(2)**

2x + 3x + 4x = 180°, so 9x = 180° and x = 20°. (1)

The largest angle is 4x = **80°**. (1)

9.3 Triangle DEF is similar to triangle ABC. The vertices are written in matching order, so D matches A, E matches B and F matches C. AB = 6 cm, BC = 8 cm and AC = 10 cm. DE = 9 cm. Calculate EF. **(2)**

Scale factor = DE / AB = 9 / 6 = 3/2. (1)

EF = 8 × 3/2 = **12 cm**. (1)

Follow through: if the scale factor is inverted but then used consistently, award the method mark only.

QUESTION 10 · Pythagoras, circles and volume · 10 marks

10.1 In a right-angled triangle the right angle lies between two sides of length 20 cm and 21 cm. Calculate the length of the hypotenuse. **(3)**

$c^2 = 20^2 + 21^2 = 400 + 441 = 841$. (1)

$c = \sqrt{841}$. (1)

c = **29 cm**. (1). The unit is part of the final mark. $29^2 = 841$.

10.2 A circle has a diameter of 20 cm. Use π = 3,14. **(4)**

10.2.1 Calculate the circumference. (2)

10.2.2 Calculate the area. (2)

The radius is 20 ÷ 2 = 10 cm. This halving is required in both parts.

10.2.1 C = 2 × 3,14 × 10 = **62,8 cm**. One mark for using r = 10 in the circumference formula, one mark for 62,8 cm.

10.2.2 A = 3,14 × 10^2 = **314 cm²**. One mark for πr^2 with r = 10, one mark for 314 cm^2.

If the diameter was used as the radius, award no marks for 10.2.1. For 10.2.2 award 1 method mark only if $A = \pi r^2$ was used with that same incorrect radius.

10.3 A closed cylinder has a radius of 7 cm and a height of 10 cm. Use π = 22/7. Calculate the volume. **(3)**

$V = \pi r^2 h$. (1)

V = 22/7 × 7 × 7 × 10. (1)

V = **1 540 cm³**. (1)

QUESTION 11 · Transformations · 4 marks

11.1 Point P(−4 ; 3) is given. Write the coordinates of the image of P after each transformation. **(4)**

11.1.1 Reflection in the y-axis. (1)

11.1.2 Reflection in the line y = x. (1)

11.1.3 A translation of 2 units right and 5 units down. (2)

11.1.1 **(4 ; 3)**. The x-coordinate changes sign. (1)

11.1.2 **(3 ; −4)**. The coordinates swap. (1)

11.1.3 (−4 + 2 ; 3 − 5) = **(−2 ; −2)**. One mark for each correct coordinate.

QUESTION 12 · Data and probability · 10 marks

Optional question. Mark it out of 10, separately from the 100.

12.1 The ages, in years, of nine players are 12, 14, 14, 15, 15, 15, 16, 17 and 18. **(5)**

12.1.1 Determine the median. (1)
12.1.2 Determine the mode. (1)
12.1.3 Determine the mean. Give the exact answer, or round it to two decimal places. (2)
12.1.4 Determine the range. (1)

The list is already sorted.

12.1.1 The middle, fifth, value is **15**. (1)

12.1.2 15 appears three times, more than any other age, so the mode is **15**. (1)

12.1.3 The sum is 136. (1) Mean = 136 ÷ 9 = **136/9**, or 15,11 years to two decimal places. (1)

12.1.4 18 – 12 = **6**. (1)

12.2 A spinner is divided into 8 equal sectors: 3 yellow, 3 red and 2 blue. **(5)**

12.2.1 Write P(red) as a simplified fraction. (2)
12.2.2 Write P(not yellow) as a simplified fraction. (2)
12.2.3 The spinner is spun 80 times. How many times do you expect blue? (1)

12.2.1 P(red) = 3/8. (2). Award 1 mark for 3/8 unsimplified if they wrote an equivalent such as 3 ÷ 8, and the second mark is already included because 3/8 is simplified.

12.2.2 Outcomes that are not yellow: 3 + 2 = 5, so P(not yellow) = **5/8**. One mark for 5 outcomes, one mark for the fraction.

12.2.3 P(blue) = 2/8 = 1/4, so the expected number is 1/4 × 80 = **20**. (1)

After marking, redo only the parts that lost marks. A second full paper the same night is less useful than fixing the three weakest questions.